6 **(a)** **(i)** All collisions are perfectly elastic. [1]

(ii) Change in momentum of molecule after colliding with wall = $2mc_x$ [1]

Time between collision with the same wall = $\frac{2L}{c_x}$,

hence, by Newton's 2nd Law, force by wall on molecule = $\frac{2mc_x}{\frac{2L}{c_x}} = \frac{mc_x^2}{L}$ [1]

By Newton's 3rd Law, the molecule exerts this force on the wall and the pressure is $\frac{\text{Force on wall}}{\text{Area of wall}} = \frac{mc_x^2}{L^3}$ [1]

Extending to N molecules each with different speeds, total pressure

$$P = \frac{Nm\langle c_x^2\rangle}{L^3}$$ [1]

Since the molecules move randomly, $\langle c^2\rangle = \langle c_x^2\rangle + \langle c_y^2\rangle + \langle c_z^2\rangle = 3\langle c_x^2\rangle$,

$$P = \frac{Nm\langle c^2\rangle}{3L^3} = \frac{M\langle c^2_x\rangle}{3V}$$ where M = total mass, V = volume of cube. [1]

Hence, $P = \frac{1}{3}\rho\langle c^2\rangle$

(b) **(i)** It is the amount of heat required per unit mass to change a substance from the liquid phase to the gaseous phase without any change in temperature. [1]

(ii) **1.** Work done by the gas $= P\,\Delta V$

$= 1.0 \times 10^5 \left[\frac{0.35}{1.6} - \frac{0.35}{790}\right]$ [1]

$= +\,2.18 \times 10^4$ J (must be positive) [1]

2. $Q = ml_v$

$= (0.35)(0.95 \times 10^6)$

$= 3.33 \times 10^5$ J [1]

3. $\Delta U = Q + W$

$= 3.3325 \times 10^5 + (-\,2.18 \times 10^4)$ [1]

$= 3.11 \times 10^5$ J [1]

{For **3.**, suggest that if '-ve' not used for W, and the answer given is 3.54×10^5 J, one mark be awarded, ie. penalise only for wrong sign for W.}

(c) **(i)** $P = \frac{m}{t}c\Delta T + h$, where h is the rate of heat loss

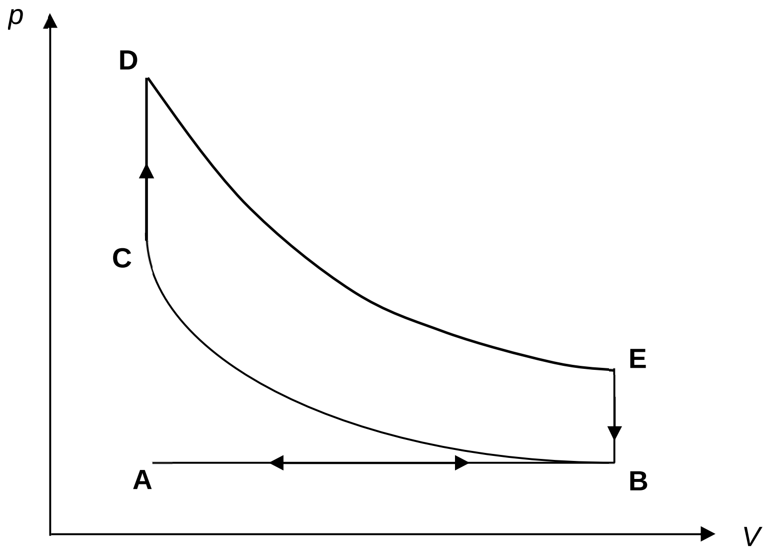
$25 = \frac{0.15}{60}c\,(19 - 15) + h$ ……………….(1)

$37 = \frac{0.23}{60}c\,(19 - 15) + h$ ……………….(2) [1]

$h = 2.50$ W [1]

(ii) The rate of heat lost would be different for each experiment causing error to the specific heat capacity calculated. [1]

(d) **(i)** **C** → **D** and **E** → **B:** Both vertical lines [1]
D → **E:** Downward curve [1]
such that **EB** is shorter than **CD** [1]



(ii) The area of the enclosed loop multiplied with the rate of the cycle (cycles per second). [1]

(iii) The area of the enclosed loop will be smaller (although the shape of the graph is unchanged). [1]